

Robotix - Inventory Turnover Analysis

Q2 2024 Report

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Distribution: Executive Team, Board of Directors

Executive Summary

Our Q2 2024 inventory analysis reveals an overall turnover ratio of 3.35x, slightly below the industry benchmark of 3.5-4.0x. Total inventory value stands at \$7.2M, with Days Inventory Outstanding (DIO) at 109.0 days. While showing improvement from Q1 (3.1x), we still have opportunities to optimize working capital efficiency.

Key Metrics Overview:

Overall Turnover Ratio: 3.35x
DIO: 109.0 days
Total Inventory Value: \$7.2M
Working Capital Impact: \$2.1M improvement potential

Inventory Turnover by Category

Product Category	Turnover Ratio	Avg. Inventory Value	DOI (Days)
Industrial Robots	4.2x	\$2.8M	86.9
Collaborative Robots	3.1x	\$1.9M	117.7
Mobile Robots	2.8x	\$1.4M	130.4
Components & Accessories	3.9x	\$1.1M	93.6

Detailed Analysis

Raw Materials Inventory: - Total Value: \$2.4M - Turnover: 4.1x - Key Components: - Motors: \$820K - Controllers: \$640K - Sensors: \$520K - Frames: \$420K

Work-in-Progress (WIP):

Total WIP Value: \$1.8M
Average Assembly Time: 12 days
Aging Profile:

- 0-30 days: 65%
- 31-60 days: 25%
- 60+ days: 10%

Finished Goods: | Product Line | Value | % of Total | |-----|-----|-----| | Industrial Series | \$1.6M | 53.3% | | Collaborative Series | \$0.8M | 26.7% | | Mobile Series | \$0.6M | 20.0% |

Location Analysis

Minneapolis Manufacturing: - Inventory Value: \$4.8M - Turnover: 3.8x - Utilization: 82%

Portland Distribution: - Inventory Value: \$2.4M - Turnover: 2.9x - Utilization: 71%

Quarter-over-Quarter Comparison

Metric	Q1 2024	Q2 2024	Change
Turnover Ratio	3.1x	3.35x	+8.1%
Total Inventory	\$7.8M	\$7.2M	-7.7%
DIO	117.7	109.0	-7.4%

Risk Factors

- Obsolescence Risk:** - Industrial Series controllers: \$180K at risk - Legacy sensor modules: \$90K exposure
- Supply Chain:** - Semiconductor lead times: 16-20 weeks - Critical component stockouts: 3 incidents in Q2

Recommendations

- Inventory Optimization:** - Implement min-max levels for top 20 components - Reduce safety stock for stable demand items - Target: Release \$2.1M working capital by Q4
- Process Improvements:** - Enhanced demand forecasting accuracy - Vendor managed inventory for non-critical parts - Cross-dock operations for distribution center

Action Items

- Immediate Actions (Next 30 Days):** - Clear aged WIP inventory (\$220K value) - Implement ABC analysis for component ordering - Review slow-moving SKUs (>180 days)
- Medium Term (60-90 Days):** - Deploy inventory optimization software - Negotiate vendor lead times - Establish KPI dashboard for real-time monitoring

Working Capital Impact

Current Working Capital Tied in Inventory: \$7.2M
Target Working Capital (Q4 2024): \$5.1M
Potential Release: \$2.1M

Cash Conversion Cycle:
- DIO: 109.0 days
- DSO: 45.0 days
- DPO: 60.0 days
Net CCC: 94.0 days

Conclusion

While showing improvement from Q1, our inventory management still requires attention to reach industry benchmarks. Implementation of recommended actions could release significant working capital while maintaining service levels. Key focus areas remain WIP reduction and finished goods optimization.

Next Review: July 31, 2024

Michael Rodriguez
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Robotix